Solar Fe I appendix

Sun · Fe I · downloadable appendix

Document	Solar Fe I appendix
Subject	Solar photospheric abundance appendix · Fe I
Author	Ryan Schmitt
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Fe.json v1.531 · 183 Fe I products · 233 total

Solar Fe I anchor: 7.477 ± 0.037

HARPS · Reference Grade · Amarsi 3D-NLTE · n = 21. The selected Fe I product anchors this Solar appendix; products and engines are not averaged together.

Highlighted band products

near-UV

7.452 ± 0.056

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · n = 51

Uncertainty budget closed for six of eight rows (RYA-1226); two Fe I Kitt Peak/molecfit rows keep a legacy bar pending their microturbulence bracket. On the migrated rows most of the quoted systematic is a conservative profile-width term carried from a measurement in another band.

VIS

7.504 ± 0.034

Kitt Peak Molecfit · Synth · 3D-NLTE · Amarsi · Reference Grade · n = 21

VIS

7.477 ± 0.037

HARPS · Synth · 3D-NLTE · Amarsi · Reference Grade · n = 21

red-optical

7.440 ± 0.033

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · n = 97

NIR

7.481 ± 0.034

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · n = 1

H

7.649 ± 0.042

CRIRES+ · Synth · 1D-NLTE · Gerber · Codex Grade · n = 8

Error-bar forest

How to read this forest. Each measured row is a separate abundance product for an instrument, holding, line pool and treatment; engines are never averaged together. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Green and gold regions mark the Asplund and Lodders literature values and uncertainties, not pass/fail gates. Grades describe the line pool's provenance and selection, not the quality of an engine. Reddish-orange mean-3D / Frankenstein rows are experimental and not adopted. Unmeasured model rows remain N/A.

Grades and model labels

Reference Grade. An externally published reference line pool, using that reference's line selection and oscillator strengths; the current Fe reference products use AGSS21. This is a provenance category, not a promise that every line has an individual primary-laboratory uncertainty.

Codex Grade. The vetted Codex pool of primary-laboratory oscillator strengths, restricted to lines at or below the adopted feature-depth gate.

Deep Grade. The corresponding laboratory-data pool above that depth gate. These stronger lines extend coverage but require particular care with saturation and blends.

1D-LTE uses a one-dimensional atmosphere with atomic populations in local thermodynamic equilibrium. **1D-NLTE** allows departures from that equilibrium; the named implementation supplies the correction or synthesis treatment.

3D-NLTE (Amarsi) includes three-dimensional atmospheric structure and non-LTE line formation; these products apply the published Amarsi correction grid to a 1D baseline.

Mean-3D / Frankenstein combines a spatially and temporally averaged 3D atmosphere with the Gerber synthesis route. Averaging removes the full 3D fluctuations and velocity field; this experimental hybrid is not a full 3D calculation or an adopted engine.

Synthesis fits the spectral line profile. **EW** infers abundance from a line's integrated absorption (equivalent width). See Methodology for the full model, line-selection and uncertainty procedures.

Asplund, Amarsi & Grevesse 2021 7.46 ± 0.04 · Lodders+ 2025 7.49 ± 0.01

bar = σ as Asplund 2021 quote it (scatter, 3D, NLTE) · blue = regular model result · reddish orange = experimental Frankenstein / Gerber mean-3D; not adopted · green = Asplund; gold = Lodders

near-UV — Kitt Peak — Kurucz 2005 corrected Codex deep pool
solar_kpno_kurucz2005_corrected · DEEPGRADED (no graded product in this band)
A(Fe) dex; axis 7.28–7.77

Synth · 1D-LTE Deep Grade · n=55		7.554 ±0.068

near-UV — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.28–7.77

Synth · 1D-LTE Reference Grade · n=48		7.515 ±0.067
Synth · 1D-LTE · Gerber Reference Grade · n=51		7.494 ±0.058
Synth · 1D-NLTE · Bergemann Reference Grade · n=48		7.523 ±0.067

near-UV — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.28–7.77

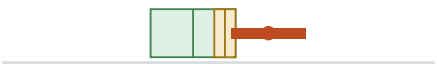
Synth · 1D-NLTE · Gerber Reference Grade · n=51		7.559 ±0.058
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=52		7.588 ±0.062
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=53		7.678 ±0.060

near-UV — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.28–7.77

Synth · 1D-LTE Reference Grade · n=46		7.380 ±0.071
Synth · 1D-LTE · Gerber Reference Grade · n=50		7.398 ±0.056
Synth · 1D-NLTE · Bergemann Reference Grade · n=46		7.389 ±0.071
Synth · 1D-NLTE · Gerber Reference Grade · n=51		7.452 ±0.056
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=51		7.500 ±0.058
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=51		7.543 ±0.053

near-UV axis: 7.28 · 7.40 · 7.52 · 7.65 · 7.77 A(Fe) dex

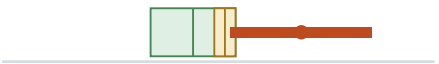
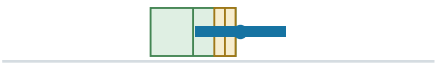
VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.28–7.69

Synth · 1D-LTE Reference Grade · n=22		7.443 ±0.034
Synth · 1D-LTE · Gerber Reference Grade · n=22		7.450 ±0.034
Synth · 1D-NLTE · Bergemann Reference Grade · n=18		7.462 ±0.035
Synth · 1D-NLTE · Gerber Reference Grade · n=22		7.484 ±0.035
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.531 ±0.035

VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.28–7.69

Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.546 ±0.036
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=21		7.471 ±0.033

VIS — Kitt Peak — Kurucz 2005 corrected Codex graded pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.28–7.69

EW · 1D-LTE Codex Grade · n=13 · 1 alternate		7.445 ±0.104
Synth · 1D-LTE Codex Grade · n=65		7.410 ±0.040
Synth · 1D-LTE · Gerber Codex Grade · n=67		7.439 ±0.067
EW · 1D-NLTE · Bergemann Codex Grade · n=12		7.455 ±0.107
Synth · 1D-NLTE · Bergemann Codex Grade · n=59		7.411 ±0.041
Synth · 1D-NLTE · Gerber Codex Grade · n=67		7.488 ±0.066
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=67		7.541 ±0.067
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=67		7.562 ±0.066
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.505 ±0.043

VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.28–7.69

Synth · 1D-LTE Reference Grade · n=166		7.367 ±0.034
Synth · 1D-LTE · Gerber Reference Grade · n=167		7.373 ±0.034
Synth · 1D-NLTE · Bergemann Reference Grade · n=156		7.371 ±0.034
Synth · 1D-NLTE · Gerber Reference Grade · n=167		7.430 ±0.034

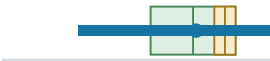
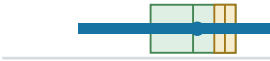
VIS — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.28–7.69

Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=101		7.485 ±0.041

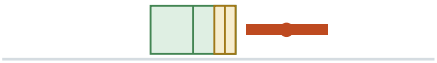
VIS — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.28–7.69

Synth · 1D-LTE Reference Grade · n=22		7.455 ±0.034
Synth · 1D-LTE · Gerber Reference Grade · n=22		7.462 ±0.034
Synth · 1D-NLTE · Bergemann Reference Grade · n=18		7.469 ±0.036
Synth · 1D-NLTE · Gerber Reference Grade · n=22		7.495 ±0.035
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.542 ±0.035
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.562 ±0.036
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=21		7.504 ±0.034

VIS — Kitt Peak — 1984 Molecfit corrected Codex graded pool
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.28–7.69

EW · 1D-LTE Codex Grade · n=13		7.463 ±0.111
Synth · 1D-LTE Codex Grade · n=65		7.414 ±0.039
Synth · 1D-LTE · Gerber Codex Grade · n=65		7.415 ±0.038
EW · 1D-NLTE · Bergemann Codex Grade · n=12		7.464 ±0.112
Synth · 1D-NLTE · Bergemann Codex Grade · n=59		7.420 ±0.040
Synth · 1D-NLTE · Gerber Codex Grade · n=65		7.467 ±0.038
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=65		7.511 ±0.038

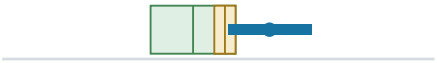
VIS — Kitt Peak — 1984 Molecfit corrected Codex graded pool solar_kpno_molecfit_corrected A(Fe) dex; axis 7.28–7.69

Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Codex Grade · n=65		7.548 ±0.038
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.512 ±0.043

VIS — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool solar_kpno_molecfit_corrected A(Fe) dex; axis 7.28–7.69

Synth · 1D-LTE Reference Grade · n=167		7.368 ±0.034
Synth · 1D-LTE · Gerber Reference Grade · n=166		7.378 ±0.034
Synth · 1D-NLTE · Bergemann Reference Grade · n=157		7.377 ±0.034
Synth · 1D-NLTE · Gerber Reference Grade · n=167		7.432 ±0.033
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=101		7.492 ±0.041

VIS — HARPS Reference Grade · Asplund replication solar_harps_molecfit_corrected A(Fe) dex; axis 7.28–7.69

Synth · 1D-LTE Reference Grade · n=22		7.486 ±0.040
Synth · 1D-LTE · Gerber Reference Grade · n=22		7.491 ±0.039
Synth · 1D-NLTE · Bergemann Reference Grade · n=18		7.523 ±0.042
Synth · 1D-NLTE · Gerber Reference Grade · n=22		7.532 ±0.039
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.577 ±0.041
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.617 ±0.040
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=21		7.477 ±0.037

VIS — HARPS Codex graded pool solar_harps_molecfit_corrected
A(Fe) dex; axis 7.28–7.69

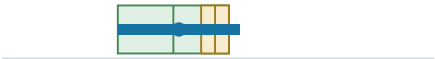
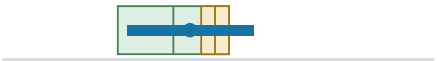
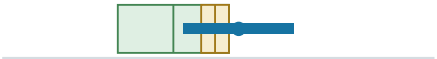
EW · 1D-LTE Codex Grade · n=6 · 1 alternate		7.498 ±0.120
Synth · 1D-LTE Codex Grade · n=171		7.428 ±0.033
Synth · 1D-LTE · Gerber Codex Grade · n=171		7.436 ±0.033
EW · 1D-NLTE · Bergemann Codex Grade · n=5		7.433 ±0.123
Synth · 1D-NLTE · Bergemann Codex Grade · n=144		7.429 ±0.034
Synth · 1D-NLTE · Gerber Codex Grade · n=171		7.492 ±0.033
Synth · 3D-NLTE · Amarsi · lab-gf Codex Grade · n=50		7.512 ±0.043

VIS — HARPS Reference Grade · full lab pool solar_harps_molecfit_corrected
A(Fe) dex; axis 7.28–7.69

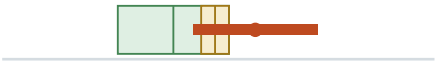
Synth · 1D-LTE Reference Grade · n=167		7.428 ±0.036
Synth · 1D-LTE · Gerber Reference Grade · n=167		7.441 ±0.035
Synth · 1D-NLTE · Bergemann Reference Grade · n=157		7.439 ±0.036
Synth · 1D-NLTE · Gerber Reference Grade · n=167		7.491 ±0.035
Synth · 3D-NLTE · Amarsi · lab-gf Reference Grade · n=100		7.500 ±0.042

VIS axis: 7.28 · 7.38 · 7.48 · 7.59 · 7.69 A(Fe) dex

red-optical — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.34–7.65

Synth · 1D-LTE Reference Grade · n=11		7.460 ±0.044
Synth · 1D-LTE · Gerber Reference Grade · n=11		7.464 ±0.044
Synth · 1D-NLTE · Bergemann Reference Grade · n=4		7.472 ±0.046
Synth · 1D-NLTE · Gerber Reference Grade · n=11		7.507 ±0.040

red-optical — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.34–7.65

Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.519 ±0.045
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.562 ±0.041

red-optical — Kitt Peak — Kurucz 2005 corrected Codex graded pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.34–7.65

Synth · 1D-LTE Codex Grade · n=97		7.437 ±0.033
Synth · 1D-LTE · Gerber Codex Grade · n=97		7.440 ±0.033
Synth · 1D-NLTE · Bergemann Codex Grade · n=46		7.449 ±0.034
Synth · 1D-NLTE · Gerber Codex Grade · n=97		7.487 ±0.033

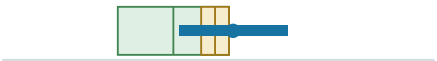
red-optical — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.34–7.65

Synth · 1D-LTE Reference Grade · n=68		7.423 ±0.034
Synth · 1D-LTE · Gerber Reference Grade · n=66		7.421 ±0.034
Synth · 1D-NLTE · Bergemann Reference Grade · n=34		7.418 ±0.035
Synth · 1D-NLTE · Gerber Reference Grade · n=68		7.455 ±0.034

red-optical — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.34–7.65

Synth · 1D-LTE Reference Grade · n=11		7.449 ±0.043
Synth · 1D-LTE · Gerber Reference Grade · n=11		7.451 ±0.043
Synth · 1D-NLTE · Bergemann Reference Grade · n=4		7.488 ±0.048

red-optical — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.34–7.65

Synth · 1D-NLTE · Gerber Reference Grade · n=11		7.503 ±0.039
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.525 ±0.044
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.554 ±0.041

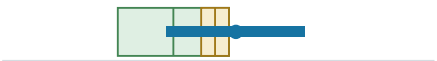
red-optical — Kitt Peak — 1984 Molecfit corrected Codex graded pool solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.34–7.65

Synth · 1D-LTE Codex Grade · n=66		7.413 ±0.034
Synth · 1D-LTE · Gerber Codex Grade · n=67		7.419 ±0.034
Synth · 1D-NLTE · Bergemann Codex Grade · n=33		7.403 ±0.036
Synth · 1D-NLTE · Gerber Codex Grade · n=65		7.454 ±0.035

red-optical — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.34–7.65

Synth · 1D-LTE Reference Grade · n=66		7.413 ±0.034
Synth · 1D-LTE · Gerber Reference Grade · n=67		7.419 ±0.034
Synth · 1D-NLTE · Bergemann Reference Grade · n=33		7.403 ±0.036
Synth · 1D-NLTE · Gerber Reference Grade · n=65		7.454 ±0.035

red-optical — IAG Reference Grade · Asplund replication solar_iag
A(Fe) dex; axis 7.34–7.65

Synth · 1D-LTE Reference Grade · n=11		7.477 ±0.044
Synth · 1D-LTE · Gerber Reference Grade · n=11		7.480 ±0.044
Synth · 1D-NLTE · Bergemann Reference Grade · n=4		7.505 ±0.050

red-optical — IAG Reference Grade · Asplund replication solar_iag
A(Fe) dex; axis 7.34–7.65

Synth · 1D-NLTE · Gerber Reference Grade · n=11		7.522 ±0.041
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.545 ±0.045
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=11		7.576 ±0.042

red-optical — IAG Codex graded pool solar_iag
A(Fe) dex; axis 7.34–7.65

Synth · 1D-LTE Codex Grade · n=96		7.437 ±0.036
Synth · 1D-LTE · Gerber Codex Grade · n=96		7.437 ±0.036
Synth · 1D-NLTE · Bergemann Codex Grade · n=46		7.446 ±0.045
Synth · 1D-NLTE · Gerber Codex Grade · n=95		7.479 ±0.036

red-optical — IAG Reference Grade · full lab pool solar_iag
A(Fe) dex; axis 7.34–7.65

Synth · 1D-LTE Reference Grade · n=67		7.424 ±0.039
Synth · 1D-LTE · Gerber Reference Grade · n=66		7.423 ±0.039
Synth · 1D-NLTE · Bergemann Reference Grade · n=33		7.422 ±0.052
Synth · 1D-NLTE · Gerber Reference Grade · n=66		7.454 ±0.039

red-optical axis: 7.34 · 7.41 · 7.49 · 7.57 · 7.65 A(Fe) dex

NIR — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Reference Grade · n=1		7.478 ±0.034
Synth · 1D-LTE · Gerber Reference Grade · n=1		7.481 ±0.034
Synth · 1D-NLTE · Gerber Reference Grade · n=1		7.525 ±0.034

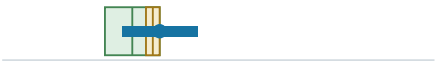
NIR — Kitt Peak — Kurucz 2005 corrected Reference Grade · Asplund replication
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.27–7.90

Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.533 ±0.035
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.569 ±0.035

NIR — Kitt Peak — Kurucz 2005 corrected Codex graded pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Codex Grade · n=21		7.517 ±0.055
Synth · 1D-LTE · Gerber Codex Grade · n=21		7.508 ±0.055
Synth · 1D-NLTE · Bergemann Codex Grade · n=5		7.538 ±0.093

NIR — Kitt Peak — Kurucz 2005 corrected Reference Grade · full lab pool
solar_kpno_kurucz2005_corrected
A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Reference Grade · n=21		7.517 ±0.055
Synth · 1D-LTE · Gerber Reference Grade · n=21		7.500 ±0.055
Synth · 1D-NLTE · Bergemann Reference Grade · n=5		7.538 ±0.093
Synth · 1D-NLTE · Gerber Reference Grade · n=19		7.552 ±0.055
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=22		7.586 ±0.055
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=20		7.622 ±0.054

NIR — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication
solar_kpno_molecfit_corrected
A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Reference Grade · n=1		7.451 ±0.043
Synth · 1D-LTE · Gerber Reference Grade · n=1		7.454 ±0.041

NIR — Kitt Peak — 1984 Molecfit corrected Reference Grade · Asplund replication solar_kpno_molecfit_corrected A(Fe) dex; axis 7.27–7.90

Synth · 1D-NLTE · Gerber Reference Grade · n=1		7.498 ±0.041
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.506 ±0.041
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.541 ±0.041

NIR — Kitt Peak — 1984 Molecfit corrected Codex graded pool solar_kpno_molecfit_corrected A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Codex Grade · n=18		7.495 ±0.047
Synth · 1D-LTE · Gerber Codex Grade · n=20		7.482 ±0.045
Synth · 1D-NLTE · Bergemann Codex Grade · n=5		7.368 ±0.068

NIR — Kitt Peak — 1984 Molecfit corrected Reference Grade · full lab pool solar_kpno_molecfit_corrected A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Reference Grade · n=18		7.495 ±0.047
Synth · 1D-LTE · Gerber Reference Grade · n=20		7.478 ±0.059
Synth · 1D-NLTE · Bergemann Reference Grade · n=5		7.368 ±0.068
Synth · 1D-NLTE · Gerber Reference Grade · n=20		7.514 ±0.060
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=20		7.559 ±0.059
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=20		7.577 ±0.060

NIR — IAG Reference Grade · Asplund replication solar_iag A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Reference Grade · n=1		7.467 ±0.035

NIR — IAG Reference Grade · Asplund replication solar_iag
A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE · Gerber Reference Grade · n=1		7.471 ±0.035
Synth · 1D-NLTE · Gerber Reference Grade · n=1		7.514 ±0.035
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.523 ±0.036
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=1		7.558 ±0.036

NIR — IAG Codex graded pool solar_iag
A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Codex Grade · n=24		7.460 ±0.075
Synth · 1D-LTE · Gerber Codex Grade · n=23		7.468 ±0.078
Synth · 1D-NLTE · Bergemann Codex Grade · n=7		7.369 ±0.069

NIR — IAG Reference Grade · full lab pool solar_iag
A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Reference Grade · n=17		7.520 ±0.099
Synth · 1D-LTE · Gerber Reference Grade · n=17		7.493 ±0.099
Synth · 1D-NLTE · Bergemann Reference Grade · n=4		7.515 ±0.112
Synth · 1D-NLTE · Gerber Reference Grade · n=17		7.555 ±0.099
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=17		7.548 ±0.098
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=17		7.598 ±0.098

NIR — CRIRES+ Codex graded pool solar_crires_plus_y_wide_rya1054
A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Codex Grade · n=5		7.551 ±0.093

NIR — CRIRES+ Codex graded pool solar_crires_plus_y_wide_rya1054
A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE · Gerber Codex Grade · n=5		7.552 ±0.092
Synth · 1D-NLTE · Bergemann Codex Grade · n=2		7.637 ±0.235

NIR — CRIRES+ Reference Grade · full lab pool solar_crires_plus_y_wide_rya1054
A(Fe) dex; axis 7.27–7.90

Synth · 1D-LTE Reference Grade · n=5		7.551 ±0.093
Synth · 1D-LTE · Gerber Reference Grade · n=5		7.557 ±0.085
Synth · 1D-NLTE · Bergemann Reference Grade · n=2		7.637 ±0.235
Synth · 1D-NLTE · Gerber Reference Grade · n=5		7.594 ±0.096
Synth · <3D>-LTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=5		7.607 ±0.080
Synth · <3D>-NLTE · Gerber · stagger · Frankenstein (experimental) Reference Grade · n=5		7.635 ±0.094

NIR axis: 7.27 · 7.43 · 7.59 · 7.74 · 7.90 A(Fe) dex

H — CRIRES+ Codex graded pool solar_crires_plus_h_rya1232
A(Fe) dex; axis 7.39–7.72

Synth · 1D-LTE Codex Grade · n=19		7.497 ±0.046
Synth · 1D-LTE · Gerber Codex Grade · n=19		7.502 ±0.046
Synth · 1D-NLTE · Gerber Codex Grade · n=8		7.649 ±0.042

H — CRIRES+ Reference Grade · full lab pool solar_crires_plus_h_rya1232
A(Fe) dex; axis 7.39–7.72

Synth · 1D-LTE Reference Grade · n=19		7.497 ±0.046
Synth · 1D-LTE · Gerber Reference Grade · n=19		7.502 ±0.046
Synth · 1D-NLTE · Gerber Reference Grade · n=8		7.649 ±0.042

H axis: 7.39 · 7.47 · 7.56 · 7.64 · 7.72 A(Fe) dex

Read all product values, including alternate line sets and experiments

CRIRES+ · Synth · 1D-LTE · Codex Grade · solar_crires_plus_h_rya1232 · GRADED · SYNTH · 1D-LTE: 7.497 ± 0.046 (Asplund 2021 convention); n=19; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Gerber · Codex Grade · solar_crires_plus_h_rya1232 · GRADED · SYNTH · ENGINE-B-NLTE: 7.649 ± 0.042 (Asplund 2021 convention); n=8; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · solar_crires_plus_h_rya1232 · GRADED · SYNTH · synth-1D-LTE-gerber: 7.502 ± 0.046 (Asplund 2021 convention); n=19; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Reference Grade · solar_crires_plus_h_rya1232 · REFERENCE · SYNTH · 1D-LTE: 7.497 ± 0.046 (Asplund 2021 convention); n=19; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Gerber · Reference Grade · solar_crires_plus_h_rya1232 · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.649 ± 0.042 (Asplund 2021 convention); n=8; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Gerber · Reference Grade · solar_crires_plus_h_rya1232 · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.502 ± 0.046 (Asplund 2021 convention); n=19; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · 1D-LTE: 7.551 ± 0.093 (Asplund 2021 convention); n=5; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · ENGINE-A: 7.637 ± 0.235 (Asplund 2021 convention); n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-LTE · Gerber · Codex Grade · solar_crires_plus_y_wide_rya1054 · GRADED · SYNTH · synth-1D-LTE-gerber: 7.552 ± 0.092 (Asplund 2021 convention); n=5; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · 1D-LTE: 7.551 ± 0.093 (Asplund 2021 convention); n=5; ξ MEASURED

CRIRES+ · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · ENGINE-A: 7.637 ± 0.235 (Asplund 2021 convention); n=2; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

CRIRES+ · Synth · 1D-NLTE · Gerber · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.594 ± 0.096 (Asplund 2021 convention); n=5; ξ MEASURED

CRIRES+ · Synth · 1D-LTE · Gerber · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.557 ± 0.085 (Asplund 2021 convention); n=5; ξ MEASURED

CRIRES+ · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.607 ± 0.080 (Asplund 2021 convention); n=5; ξ MEASURED · experimental, not adopted

CRIRES+ · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_crires_plus_y_wide_rya1054 · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.635 ± 0.094 (Asplund 2021 convention); n=5; ξ MEASURED · experimental, not adopted

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.467 ± 0.035 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.514 ± 0.035 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.471 ± 0.035 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

IAG · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.523 ± 0.036 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

IAG · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.558 ± 0.036 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

IAG · Synth · 1D-LTE · Codex Grade · solar_iag · GRADED · SYNTH · 1D-LTE: 7.460 ± 0.075 (Asplund 2021 convention); n=24; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A: 7.369 ± 0.069 (Asplund 2021 convention); n=7; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · synth-1D-LTE-gerber: 7.468 ± 0.078 (Asplund 2021 convention); n=23; ξ MEASURED

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · REFERENCE · SYNTH · 1D-LTE: 7.520 ± 0.099 (Asplund 2021 convention); n=17; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-A: 7.515 ± 0.112 (Asplund 2021 convention); n=4; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.555 ± 0.099 (Asplund 2021 convention); n=17; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.493 ± 0.099 (Asplund 2021 convention); n=17; ξ MEASURED

IAG · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.548 ± 0.098 (Asplund 2021 convention); n=17; ξ MEASURED · experimental, not adopted

IAG · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.598 ± 0.098 (Asplund 2021 convention); n=17; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.478 ± 0.034 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.525 ± 0.034 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.481 ± 0.034 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.533 ± 0.035 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.569 ± 0.035 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from

this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.517 ± 0.055 (Asplund 2021 convention); n=21; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.538 ± 0.093 (Asplund 2021 convention); n=5; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.508 ± 0.055 (Asplund 2021 convention); n=21; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.517 ± 0.055 (Asplund 2021 convention); n=21; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.538 ± 0.093 (Asplund 2021 convention); n=5; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.552 ± 0.055 (Asplund 2021 convention); n=19; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.500 ± 0.055 (Asplund 2021 convention); n=21; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.586 ± 0.055 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.622 ± 0.054 (Asplund 2021 convention); n=20; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.451 ± 0.043 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.498 ± 0.041 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.454 ± 0.041 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.506 ± 0.041 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.541 ± 0.041 (Asplund 2021 convention); n=1; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.495 ± 0.047 (Asplund 2021 convention); n=18; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.368 ± 0.068 (Asplund 2021 convention); n=5; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.482 ± 0.045 (Asplund 2021 convention); n=20; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.495 ± 0.047 (Asplund 2021 convention); n=18; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.368 ± 0.068 (Asplund 2021 convention); n=5; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.514 ± 0.060 (Asplund 2021 convention); n=20; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.478 ± 0.059 (Asplund 2021 convention); n=20; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.559 ± 0.059 (Asplund 2021 convention); n=20; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.577 ± 0.060 (Asplund 2021 convention); n=20; ξ MEASURED · experimental, not adopted

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A-3DNLTE: 7.477 ± 0.037 (Asplund 2021 convention); n=21; ξ NOT_APPLICABLE

HARPS · Synth · 1D-LTE · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.486 ± 0.040 (Asplund 2021 convention); n=22; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.523 ± 0.042 (Asplund 2021 convention); n=18; ξ MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.532 ± 0.039 (Asplund 2021 convention); n=22; ξ MEASURED

HARPS · Synth · 1D-LTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.491 ± 0.039 (Asplund 2021 convention); n=22; ξ MEASURED

HARPS · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.577 ± 0.041 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

HARPS · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_harps_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.617 ± 0.040 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

HARPS · Synth · 1D-LTE · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED-LOCALRENORM · SYNTH · 1D-LTE: 7.339 ± 0.087 (Asplund 2021 convention); n=109; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED-LOCALRENORM · SYNTH · ENGINE-A: 7.350 ± 0.088 (Asplund 2021 convention); n=105; ξ MEASURED

HARPS · Synth · 1D-LTE · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.419 ± 0.039 (Asplund 2021 convention); n=106; ξ MEASURED

HARPS · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_harps_molecfit_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.429 ± 0.040 (Asplund 2021 convention); n=102; ξ MEASURED

HARPS · EW · 1D-LTE · Codex Grade · solar_harps_molecfit_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.498 ± 0.120 (Asplund 2021 convention); n=6; ξ MEASURED

HARPS · EW · 1D-NLTE · Bergemann · Codex Grade · solar_harps_molecfit_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.433 ± 0.123 (Asplund 2021 convention); n=5; ξ MEASURED

HARPS · EW · 1D-LTE · Codex Grade · solar_harps_molecfit_corrected · GRADED · PROFILEFIT · ENGINE-B: 7.431 ± 0.122 (Asplund 2021 convention); n=5; ξ MEASURED

HARPS · Synth · 1D-LTE · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.428 ± 0.033 (Asplund 2021 convention); n=171; ξ MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.512 ± 0.043 (Asplund 2021 convention); n=50; ξ NOT_APPLICABLE

HARPS · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.429 ± 0.034 (Asplund 2021 convention); n=144; ξ MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.492 ± 0.033 (Asplund 2021 convention); n=171; ξ MEASURED

HARPS · Synth · 1D-LTE · Gerber · Codex Grade · solar_harps_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.436 ± 0.033 (Asplund 2021 convention); n=171; ξ MEASURED

HARPS · Synth · 1D-LTE · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.428 ± 0.036 (Asplund 2021 convention); n=167; ξ MEASURED

HARPS · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE: 7.500 ± 0.042 (Asplund 2021 convention); n=100; ξ NOT_APPLICABLE

HARPS · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.439 ± 0.036 (Asplund 2021 convention); n=157; ξ MEASURED

HARPS · Synth · 1D-NLTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.491 ± 0.035 (Asplund 2021 convention); n=167; ξ MEASURED

HARPS · Synth · 1D-LTE · Gerber · Reference Grade · solar_harps_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.441 ± 0.035 (Asplund 2021 convention); n=167; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A-3DNLTE: 7.471 ± 0.033 (Asplund 2021 convention); n=21; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.443 ± 0.034 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.462 ± 0.035 (Asplund 2021 convention); n=18; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.484 ± 0.035 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.450 ± 0.034 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.531 ± 0.035 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.546 ± 0.036 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.329 ± 0.036 (Asplund 2021 convention); n=106; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.340 ± 0.036 (Asplund 2021 convention); n=101; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.445 ± 0.104 (Asplund 2021 convention); n=13; ξ MEASURED

Kitt Peak · EW · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.455 ± 0.107 (Asplund 2021 convention); n=12; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · PROFILEFIT · ENGINE-B: 7.479 ± 0.090 (Asplund 2021 convention); n=9; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.410 ± 0.040 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.505 ± 0.043 (Asplund 2021 convention); n=50; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.411 ± 0.041 (Asplund 2021 convention); n=59; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.488 ± 0.066 (Asplund 2021 convention); n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.439 ± 0.067 (Asplund 2021 convention); n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND.

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.541 ± 0.067 (Asplund 2021 convention); n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.562 ± 0.066 (Asplund 2021 convention); n=67; ξ UNMEASURED · INCOMPLETE: no dA/dxi exists for this pool, so the xi term is absent from this bar rather than zero. The bar is a LOWER BOUND. · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.367 ± 0.034 (Asplund 2021 convention); n=166; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE: 7.485 ± 0.041 (Asplund 2021 convention); n=101; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.371 ± 0.034 (Asplund 2021 convention); n=156; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.430 ± 0.034 (Asplund 2021 convention); n=167; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.373 ± 0.034 (Asplund 2021 convention); n=167; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A-3DNLTE: 7.504 ± 0.034 (Asplund 2021 convention); n=21; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.455 ± 0.034 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.469 ± 0.036 (Asplund 2021 convention); n=18; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.495 ± 0.035 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.462 ± 0.034 (Asplund 2021 convention); n=22; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.542 ± 0.035 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.562 ± 0.036 (Asplund 2021 convention); n=22; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.347 ± 0.036 (Asplund 2021 convention); n=106; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Deep Grade · solar_kpno_molecfit_corrected · DEEPGRADED · SYNTH · ENGINE-A: 7.357 ± 0.037 (Asplund 2021 convention); n=102; ξ MEASURED

Kitt Peak · EW · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · PROFILEFIT · 1D-LTE: 7.463 ± 0.111 (Asplund 2021 convention); n=13; ξ MEASURED

Kitt Peak · EW · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · PROFILEFIT · ENGINE-A: 7.464 ± 0.112 (Asplund 2021 convention); n=12; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.414 ± 0.039 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A-3DNLTE: 7.512 ± 0.043 (Asplund 2021 convention); n=50; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.420 ± 0.040 (Asplund 2021 convention); n=59; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.467 ± 0.038 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.415 ± 0.038 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.511 ± 0.038 (Asplund 2021 convention); n=65; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.548 ± 0.038 (Asplund 2021 convention); n=65; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.368 ± 0.034 (Asplund 2021 convention); n=167; ξ MEASURED

Kitt Peak · Synth · 3D-NLTE · Amarsi · lab-gf · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A-3DNLTE: 7.492 ± 0.041 (Asplund 2021 convention); n=101; ξ NOT_APPLICABLE

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.377 ± 0.034 (Asplund 2021 convention); n=157; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.432 ± 0.033 (Asplund 2021 convention); n=167; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.378 ± 0.034 (Asplund 2021 convention); n=166; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Deep Grade · solar_kpno_kurucz2005_corrected · DEEPGRADED · SYNTH · 1D-LTE: 7.554 ± 0.068 (Asplund 2021 convention); n=55; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.515 ± 0.067 (Asplund 2021 convention); n=48; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.523 ± 0.067 (Asplund 2021 convention); n=48; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.559 ± 0.058 (Asplund 2021 convention); n=51; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.494 ± 0.058 (Asplund 2021 convention); n=51; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.588 ± 0.062 (Asplund 2021 convention); n=52; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.678 ± 0.060 (Asplund 2021 convention); n=53; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.380 ± 0.071 (Asplund 2021 convention); n=46; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.389 ± 0.071 (Asplund 2021 convention); n=46; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.452 ± 0.056 (Asplund 2021 convention); n=51; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.398 ± 0.056 (Asplund 2021 convention); n=50; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.500 ± 0.058 (Asplund 2021 convention); n=51; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.543 ± 0.053 (Asplund 2021 convention); n=51; ξ MEASURED · experimental, not adopted

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.477 ± 0.044 (Asplund 2021 convention); n=11; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.505 ± 0.050 (Asplund 2021 convention); n=4; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.522 ± 0.041 (Asplund 2021 convention); n=11; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.480 ± 0.044 (Asplund 2021 convention); n=11; ξ MEASURED

IAG · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.545 ± 0.045 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

IAG · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_iag · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.576 ± 0.042 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

IAG · Synth · 1D-LTE · Codex Grade · solar_iag · GRADED · SYNTH · 1D-LTE: 7.437 ± 0.036 (Asplund 2021 convention); n=96; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-A: 7.446 ± 0.045 (Asplund 2021 convention); n=46; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · ENGINE-B-NLTE: 7.479 ± 0.036 (Asplund 2021 convention); n=95; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Codex Grade · solar_iag · GRADED · SYNTH · synth-1D-LTE-gerber: 7.437 ± 0.036 (Asplund 2021 convention); n=96; ξ MEASURED

IAG · Synth · 1D-LTE · Reference Grade · solar_iag · REFERENCE · SYNTH · 1D-LTE: 7.424 ± 0.039 (Asplund 2021 convention); n=67; ξ MEASURED

IAG · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-A: 7.422 ± 0.052 (Asplund 2021 convention); n=33; ξ MEASURED

IAG · Synth · 1D-NLTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.454 ± 0.039 (Asplund 2021 convention); n=66; ξ MEASURED

IAG · Synth · 1D-LTE · Gerber · Reference Grade · solar_iag · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.423 ± 0.039 (Asplund 2021 convention); n=66; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.460 ± 0.044 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.472 ± 0.046 (Asplund 2021 convention); n=4; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.507 ± 0.040 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.464 ± 0.044 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.519 ± 0.045 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_kurucz2005_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.562 ± 0.041 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · 1D-LTE: 7.437 ± 0.033 (Asplund 2021 convention); n=97; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-A: 7.449 ± 0.034 (Asplund 2021 convention); n=46; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.487 ± 0.033 (Asplund 2021 convention); n=97; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_kurucz2005_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.440 ± 0.033 (Asplund 2021 convention); n=97; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · 1D-LTE: 7.423 ± 0.034 (Asplund 2021 convention); n=68; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-A: 7.418 ± 0.035 (Asplund 2021 convention); n=34; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.455 ± 0.034 (Asplund 2021 convention); n=68; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_kurucz2005_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.421 ± 0.034 (Asplund 2021 convention); n=66; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · 1D-LTE: 7.449 ± 0.043 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-A: 7.488 ± 0.048 (Asplund 2021 convention); n=4; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · ENGINE-B-NLTE: 7.503 ± 0.039 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-1D-LTE-gerber: 7.451 ± 0.043 (Asplund 2021 convention); n=11; ξ MEASURED

Kitt Peak · Synth · <3D>-LTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-LTE-gerber-stagger: 7.525 ± 0.044 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · <3D>-NLTE · Gerber · stagger · Reference Grade · solar_kpno_molecfit_corrected · Asplund et al. 2021 lines · SYNTH · synth-mean3D-NLTE-gerber-stagger: 7.554 ± 0.041 (Asplund 2021 convention); n=11; ξ MEASURED · experimental, not adopted

Kitt Peak · Synth · 1D-LTE · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · 1D-LTE: 7.413 ± 0.034 (Asplund 2021 convention); n=66; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-A: 7.403 ± 0.036 (Asplund 2021 convention); n=33; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · ENGINE-B-NLTE: 7.454 ± 0.035 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Codex Grade · solar_kpno_molecfit_corrected · GRADED · SYNTH · synth-1D-LTE-gerber: 7.419 ± 0.034 (Asplund 2021 convention); n=67; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · 1D-LTE: 7.413 ± 0.034 (Asplund 2021 convention); n=66; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Bergemann · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-A: 7.403 ± 0.036 (Asplund 2021 convention); n=33; ξ MEASURED

Kitt Peak · Synth · 1D-NLTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · ENGINE-B-NLTE: 7.454 ± 0.035 (Asplund 2021 convention); n=65; ξ MEASURED

Kitt Peak · Synth · 1D-LTE · Gerber · Reference Grade · solar_kpno_molecfit_corrected · REFERENCE · SYNTH · synth-1D-LTE-gerber: 7.419 ± 0.034 (Asplund 2021 convention); n=67; ξ MEASURED

Solar iron context

Fe I supplies the bulk of the neutral-iron line statistics used for the Solar abundance reference. Separate instruments, line pools and treatments test the stability of that reference; their differences remain visible.

Near-UV opacity — report and noted

Molecular opacity INCLUDED — OH A-X, NH, CH and CN violet, 6,223 VALD lines as Turbospectrum lists with the dominant isotopologue (RYA-1207). The band previously carried none: every molecular list iSpec ships begins at 400 nm. Measured PAIRED against a molecules-off run on the same code and pool, the lever is -0.047..-0.055 dex on Fe I and -0.276..-0.332 on Fe II, with chi2r median 81.8 -> 67.4 and control windows containing no molecular line bit-identical. RYA-1209 additionally curated out Fe I 3721.272 (systematic:K07 — gf attributed to a VALD3 reference that publishes no per-line uncertainty, no NIST accuracy class, so it never met the DEEPGRADED tier's lab-gf definition); the Fe I pool is 57 candidates and the budget clears at gf rung 3 (cited lab), 55/55 GF-LAB, RMS 0.0509 dex. RESIDUAL DEFICIT: the correction still UNDER-corrects — RYA-1204 measured synth/obs above 1 on 24 of 40 windows after it — so the band remains short of opacity by an amount not yet measured on this pool, and these values are a floor rather than a converged answer. sigma_syst is unchanged at 0.1122 because it is dominated by the 0.10 dex pseudo-continuum term, not by gf: the true continuum is never observed here. Not adopted for the headline. The headline is the per-band best-available product, selected from this feed by the publication generator rather than fixed as a constant: RYA-850 promoted the graded laboratory-gf pool to the PRIMARY reported value, and RYA-819 retired the earlier constructed anchor after showing its Magic-2013 correction wrong in both magnitude and shape. RYA-161: nothing calibrated to a target.

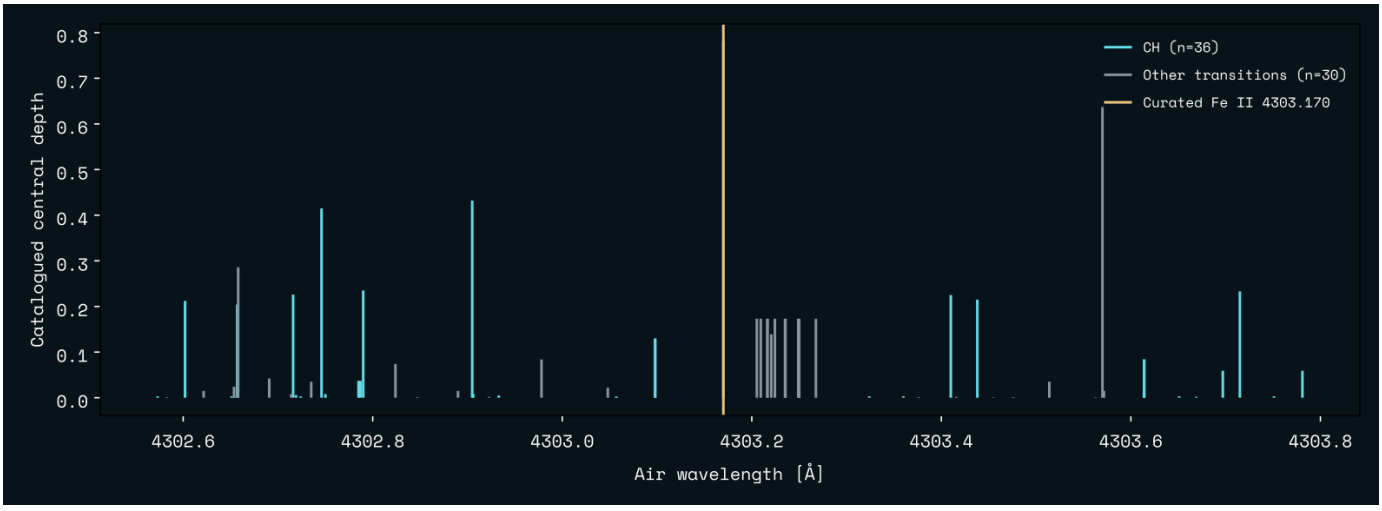
Frankenstein — experimental, pending the Bride

42 experimental products remain available in the Fe I product values and downloads.

RYA-1197 Part A: the Frankenstein mean-<3D> route is HELD FROM HEADLINE publication until the Bride validates it. It inflates A(Fe) by roughly +0.10 dex against the 1D siblings and exists on VIS/KP only, so it is not a comparable measurement of the same quantity. It stays in the feed deliberately: the LTE/NLTE mean-3D PAIR is the NLTE differential on ONE atmosphere (RYA-542), which is the diagnostic it is for. Not a headline value.

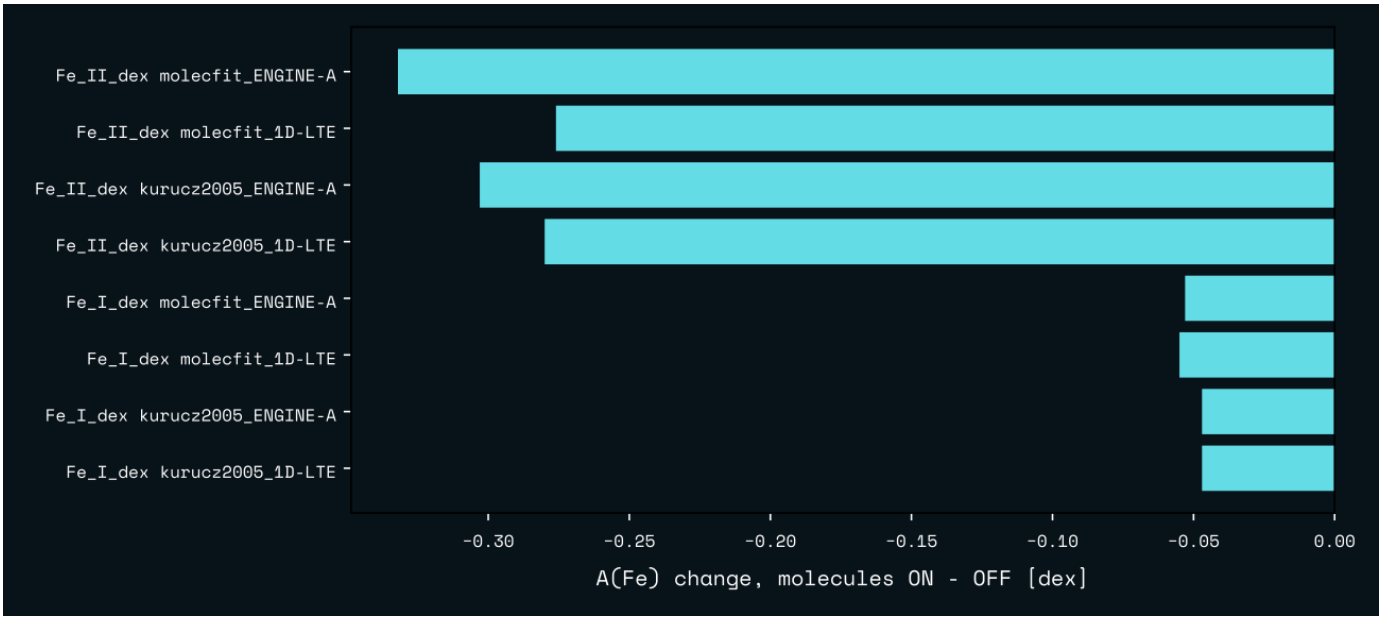
Problem-line and model diagnostics

CH G-band artifact



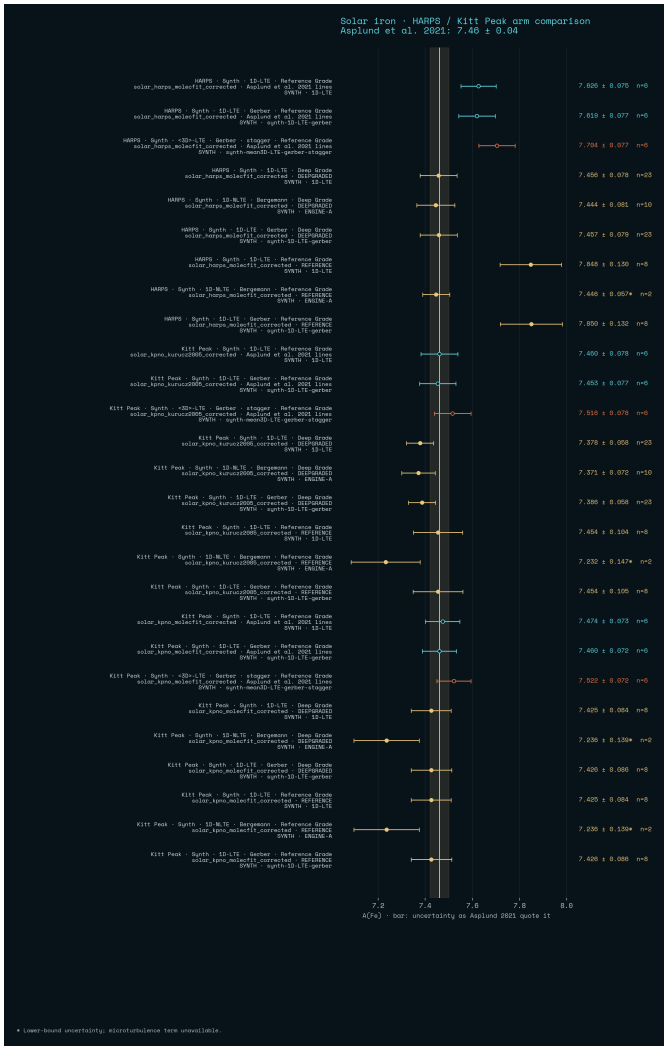
CH G-band blend artifact — the synthesis list cannot represent the dominant absorber · A = 10.0 (+2.5 dex). RYA-1191 blend audit: 66 catalogued transitions in the 0.62 A fit window, top contaminant CH I at 43.8% of the summed depth against Fe II's 11.1%. The VIS synthesis list GESv6_atom_hfs_iso carries 206 species and NOT ONE molecular, so the fit had nowhere to put the CH absorption except Fe II. RYA-515 independently counted 36 of 65 transitions as CH. The plot shows the committed transition census in the proof generator’s fit window; it is not a synthetic spectrum. Source evidence

Near-UV molecular opacity



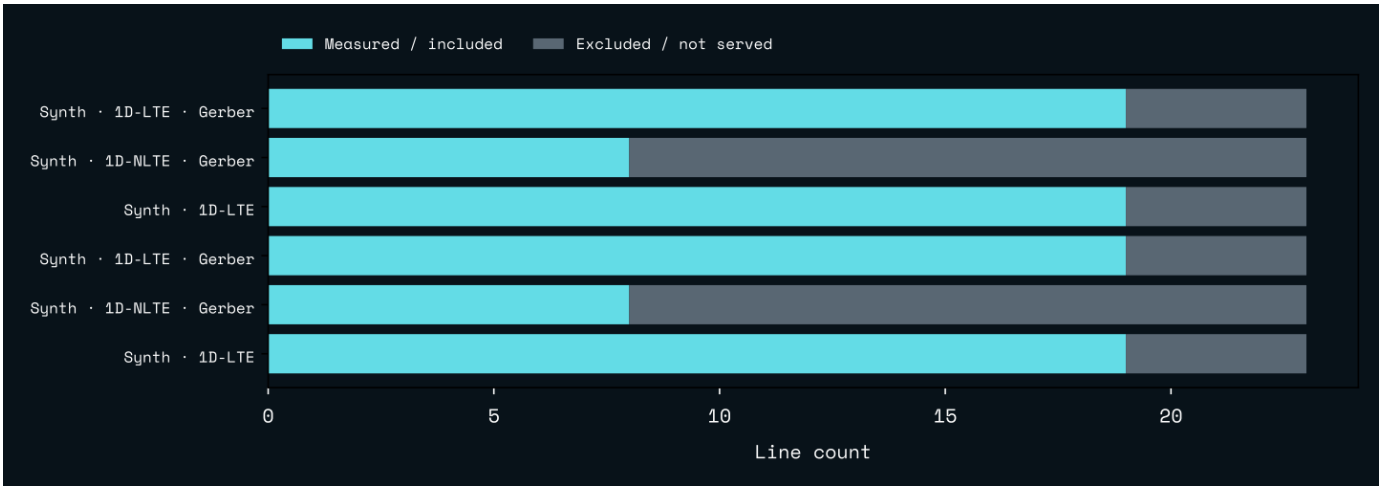
molecules OFF vs ON on THE SAME CODE and the same pool. Differencing against the published values instead would have measured the molecular lever PLUS every code change since the feed was written on Sept 3 -- which is about -0.035 dex of the total. Source evidence

HARPS arm offset



Current Fe II VIS products, separated by holding and engine. The arm offset persists after the curated CH artifact was removed. Source evidence

CRIRES+ H model coverage



Included and excluded line counts from each current product. Excluded rows are not automatically failed fits; the per-line reason distinguishes unavailable NLTE coverage. Source evidence

Download the evidence

692 graded per-line measurements exported. Evidence for 149 graded products is missing or does not reproduce the feed; 0 exported lines lack an unambiguous gf match. The download is explicitly incomplete until those upstream records are supplied.

Graded per-line abundances (CSV)

All product fields (CSV; nested fields preserved as JSON)

Per-product line-evidence coverage (CSV)

Source Fe.json

Reproducibility

Generator scripts/generate_fe_publication.py v1.0.0
Science source commit f5c6b3d35f016dd66b74d3a098459bd648f9677f
Fe.json v1.531 · feed timestamp 2026-10-29T03:00:31Z
Generated 2026-10-29T03:25:04+00:00
Feed SHA-256 9abce846505115011b45a15320e794cc4fd8bab6e9865bcd51c35635e02f4de4

Refreshed product tracker · Build manifest · Fe II appendix

References

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Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites (doi: 10.1007/s11214-025-01146-w). Table 6 present-day Solar composition, shown as the gold literature comparison.

Amarsi, A. M.; Liljegren, S.; Nissen, P. E. (2022). 3D non-LTE iron abundances in FG-type dwarfs (doi: 10.1051/0004-6361/202244542). The Amarsi 3D non-LTE iron correction grid used by the displayed products.

Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum (doi: 10.1051/0004-6361/202243673). The Gerber Turbospectrum non-LTE synthesis implementation named in the model rows.

Den Hartog, E. A.; Ruffoni, M. P.; Lawler, J. E.; Pickering, J. C.; Lind, K.; Brewer, N. R. (2014). Fe I Oscillator Strengths for Transitions from High-lying Even-parity Levels (doi: 10.1088/0067-0049/215/2/23). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Brewer, N. R.; Lind, K.; Nave, G.; Pickering, J. C. (2014). Fe I oscillator strengths for the Gaia-ESO survey (doi: 10.1093/mnras/stu780). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.

Belmonte, M. T.; Pickering, J. C.; Ruffoni, M. P.; Den Hartog, E. A.; Lawler, J. E.; Guzman, A.; Heiter, U. (2017). Fe I Oscillator Strengths for Transitions from High-lying Odd-parity Levels (doi: 10.3847/1538-4357/aa8cd3). Laboratory transition probabilities used in the resolved Fe I per-line evidence. Coverage and unresolved products are listed in the downloads.